ELEVATE LABS – DATA ANALYST INTERNSHIP

Project Report

E-commerce Return Rate Reduction Analysis

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Introduction

The e-commerce industry has experienced rapid growth, but product returns remain a major challenge. High return rates increase logistics costs, complicate inventory management, and reduce customer satisfaction. This project analyzes order history, customer behavior, product categories, and return reasons to identify the key factors behind returns and recommend strategies that improve operational efficiency and customer experience.

Project Overview

This analysis investigates patterns in historical e-commerce data to identify products with high return rates, understand customer behavior, determine common return reasons such as size mismatch, damaged products, wrong item delivery, and delayed shipping, and provide actionable recommendations through interactive dashboards.

Technologies Used

- SQL, MySQL, PostgreSQL
- Python (Pandas, NumPy)
- Tableau, Power BI, Matplotlib, Seaborn
- Scikit-learn, XGBoost
- Git & GitHub

Advantages

- Provides real-time insights into return trends.
- Supports data-driven decision-making.
- Improves inventory and logistics management.
- Enhances customer satisfaction.
- Identifies high-risk products and orders.

Limitations

- Requires accurate and complete datasets.
- Initial setup requires technical expertise.
- Large datasets may reduce dashboard performance.
- Dashboards require regular maintenance.
- Some BI tools require paid licenses.

Recommendations

Improve product descriptions, provide accurate size guides, strengthen quality control, optimize logistics, enhance packaging, and continuously monitor return trends using dashboards and predictive analytics.

Conclusion

The E-commerce Return Rate Reduction Analysis demonstrates how data analytics can reduce product returns by identifying their root causes. By leveraging SQL, Python, Tableau, and business intelligence techniques, organizations can improve operational efficiency, reduce costs, enhance customer satisfaction, and increase profitability. This project showcases the practical application of data analytics in solving real-world e-commerce challenges.

Acknowledgement

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